

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks: 40

Annexure A

CASE STUDY

Code of Conduct:

192312538

I SHAKTHI (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Shakthi

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Question 1

Case Study - Cybersecurity Intrusion Detection System

A company is designing a rule-based logical Agent to detect network intrusions. The system's knowledge base contains:

- I. If there is FailedLoginAttempts AND UnusualIPAddress $\Rightarrow$ Possible BruteForce Attack
- II. If there is FailedLoginAttempts AND OddLoginTime $\Rightarrow$ Possible Account Compromise
- III. If there is UnusualIPAddress AND HighDataTransfer $\Rightarrow$ Possible Data Exfiltration
- IV. Facts about Session X: FailedLoginAttempts = True, UnusualIPAddress = True, HighDataTransfer = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible security alerts for Session X. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm Session X indicates Data Exfiltration. Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Question 2

Case Study - Airport Priority Landing System

An airport control tower uses a FOL-based knowledge base to manage priority landing clearance.

- I. for-all x . Aircraft(x) AND EmergencyDeclared(x) $\Rightarrow$ PriorityClearance(x) [Rule 1]
- II. for-all x . PriorityClearance(x) $\Rightarrow$ RunwayReserved(x) AND NOT HoldPattern(x) [Rule 2]
- III. for-all x . for-all y . Aircraft(x) AND QueuedAfter(x, y) AND RunwayReserved(y) $\Rightarrow$ Delay(x) [Rule 3]
- IV. Facts: AircraftFlightZ, EmergencyDeclaredFlightZ, AircraftFlightY, QueuedAfterFlightY, FlightZ

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that Delay(FlightY) is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyse whether the current FOL knowledge base is sufficient to handle two simultaneous emergency aircraft. Identify what additional rules or facts are needed and justify logically.

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Question 3

Case Study - Industrial Equipment Maintenance Expert System

A factory maintenance logical agent operates with the following propositional knowledge base (KB):

- 1) P1: MachineOverheating $\Rightarrow$ MaintenanceNeeded
- 2) P2: MaintenanceNeeded AND HighPriorityLine $\Rightarrow$ ScheduleImmediateRepair
- 3) P3: NOT ScheduleImmediateRepair AND HighPriorityLine $\Rightarrow$ ProductionRisk
- 4) Facts: MachineOverheating = True, HighPriorityLine = True

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\Rightarrow$ move NOT inward $\Rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem ScheduleImmediateRepair. Negate the goal, combine with the KB in CNF and derive the empty clause. Draw the resolution proof tree.
- (c) If the fact MachineOverheating is removed from the KB, analyze step-by-step how logical conclusions change. Does ProductionRisk still follow? Apply appropriate reasoning patterns to justify.

Question 4

Case Study - Minefield Robot Knowledge Agent

An AI robot navigating a minefield clearance grid perceives: Vibration at [2,1], Smoke at [1,1], Beacon at [2,2]. The agent uses a propositional KB with these rules:

- 1) Vibration[2,1] $\Rightarrow$ Mine is adjacent to cell [2,1]
- 2) Smoke[1,1] $\Rightarrow$ Fire is adjacent to cell [1,1]
- 3) Beacon[2,2] $\Rightarrow$ SafeZoneMarked is at [2,2]
- 4) NOT Mine[x] AND NOT Fire[x] $\Rightarrow$ Safe[x]

- (a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new facts. List all derived conclusions.
- (b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Mine and Fire, each reasoning step precisely.
- (c) Evaluate whether the goal [1,1] AND [2,1] AND [2,2] is safe for the robot to reach the SafeZoneMarked. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

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Question : 1

a) Propositional logic representation :

F = failed login attempts, U = unusual IP address, O = odd login time, H = high data transfer, B = Possible Brute force attack, C = Possible account compromise, D = Possible Data Exfiltration.

The rules are represented as ;

Rule 1: $F \wedge U \rightarrow B$

Rule 2: $F \wedge O \rightarrow C$

Rule 3: $U \wedge H \rightarrow D$

The facts for session X are: F = True, U = True, H = True.

There is no fact indicating that odd login time is true.

b) Forward chaining: It starts with the known facts and applies the rules whenever their condition are satisfied.

Initial facts:

1) failed login attempts = true

2) unusual IP address = true

3) high data transfer = true

Step 1: Rule 1 states that failed login attempts, unusual IP address together indicate a possible brute force attack. Both conditions are true for session X. possible brute force attack = true.

Step 2: Rule 2 requires both failed login attempts, an odd login time. Failed login attempts are true, but there is no information showing that odd login time is true.

Step 3: Rule 3 requires an unusual IP address & high data transfer. Possible data exfiltration = true.

c) Backward chaining:

The goal is : Session X indicates data exfiltration.

Backward chaining starts with the goal & searches for a rule

that can prove it.

The rule that concludes possible data exfiltration is Rule 3. Rule 3 requires 2 conditions.

- 1) unusual IP addresses
- 2) High Data transfer

For session x: unusual IP address is true

High data transfer is true

So, both conditions required by rule 3 are satisfied.

Session x indicates possible data exfiltration.

2) Question 2:

a) verification of rule 1 with the given facts

rule 1 states: For every aircraft, if the aircraft has declared an emergency, it receives priority clearance. The rule contains the variable x. The given facts include: aircraft (flight 2)

To make both expressions identical,

Emergency declared (flight 2)

we substitute, $x = \text{flight 2}$

$\therefore$ the substitution set, on MGU, is: $\theta = \{x / \text{flight 2}\}$

After making this substitution,

~~Emergency declared~~ (flight 2)

This is exactly the same as the given fact.

$\therefore$ the rule can be applied to flight 2.

The conclusion is: Priority clearance (Flight 2)

Thus, the complete substitution is: $\theta = \{x / \text{flight 2}\}$.

Flight 2 has priority clearance.

b) Resolution proof for Delay (flight 4)

Aircraft (x) $\wedge$ Emergency Declared (x) $\rightarrow$ Priority clearance (x)

Priority clearance (x) $\rightarrow$ Runway Reserved (x) $\wedge$ Hold Pattern (x)

Priority clearance (x) $\rightarrow$ Runway Reserved (x)

Aircraft (x) $\wedge$ Arrived After (x, y) $\wedge$ Runway Reserved (y) $\rightarrow$ Delay (x)

Consult to CNF:

C1: Aircraft (Flight z) $\vee \rightarrow$ Emergency Declared (Flight z) $\vee$ Priority clearance (Flight z)

C2: Priority clearance (Flight z) $\vee$ Runway Reserved (Flight z)

C3: Aircraft (Flight y) $\vee \rightarrow$ Arrived After (Flight y, Flight z) $\vee \rightarrow$ Runway Reserved (Flight z) $\vee$ Delay (Flight y)

Resolution 1: Priority clearance (Flight z)

Resolution 2: Runway Reserved (Flight z)

Resolution 3: Delay (Flight y)

$\therefore$ Delay (Flight y) is true.

c) Two Simultaneous Emergency Aircraft:

Suppose 2 aircraft declare emergencies; Emergency declared (Flight A, B)

Problem: The current KB does not specify whether two aircraft can use the same runway simultaneously. So, it is insufficient to resolve a conflict between simultaneous emergencies.

For ex: Emergency Declared (x) $\wedge$ Emergency Declared (y) $\wedge$ Higher Priority (x, y) $\rightarrow$ Priority clearance (x)

Runway Reserved (x) $\rightarrow$ Runway Reserved (y)

where $x \neq y$ if only one runway is available.

Conclusion:

The KB needs:

1) A runway availability constraint

2) A priority comparison rule

3) A method for selecting which emergency aircraft lands first.

3) Question : 3

Let : M: machine overheating

N: Maintenance Needed

H = High priority line

S = Schedule immediate Repair

P = Production Risk

a) conversion into CNF :

P1 : original : $M \rightarrow N$

eliminate implication : $M \vee N$

It is in CNF only.

P2 : original : $N \wedge H \rightarrow S$

eliminate implication : $(N \wedge H) \vee S$

MOVE NOT inward using De Morgan's law : $N \vee H \vee S$

$\therefore$ CNF : $\neg N \vee \neg H \vee S$

P3 : original : $\neg S \wedge H \rightarrow P$

eliminate implication : $\neg(\neg S \wedge H) \vee P$

move NOT inward : $\neg(\neg S) \vee \neg H \vee P$

Double negation : $S \vee \neg H \vee P$

This is the CNF.

b) Resolution Refutation to prove S

Goal : S

Negate the goal : $\neg S$

add : C6 : $\neg S$

Resolve : C1 : $\neg M \vee N$ with C4 : M

result : $N \rightarrow$ C7 : N

Resolve : C2 : $\neg N \vee \neg H \vee S$ with C7 : N

result : $\neg H \vee S$

resolve with : $(S : H)$

result : $S \rightarrow S$ with $(b : \neg S)$

$\therefore$ Schedule Immediate Repair is proved

2) Remove the Fact Machine Overheating:

Now remove : H

remaining facts are only : H

Step 1: check P1: $H \rightarrow N$

$\therefore H$ is not known to be true, we can't derive : N

$\therefore$ Maintenance Needed can't be derived

Step 2: check P2: $N \wedge H \rightarrow S$; $H = \text{true}$; N is unknown

$\therefore$ So S can't be derived.

Step 3: check P3: $S \wedge H \rightarrow P$; $H = \text{true}$; $\neg S$

not being able to prove $S \neq$ proving $\neg S$

P can't be derived

So, Production Risk does not logically follow

4) Question : 4

Let : V_{21} = vibration at $[2, 1]$, S_{11} = smoke at $[1, 1]$, B_{22} = Beacon at $[2, 2]$,

$M_{adj}(1,1)$ = Mine is adjacent to $[2, 1]$, $F_{adj}(1,1)$ = Fire is adjacent to $[1, 1]$,

Z_{22} = Safe Zone Marker at $[2, 2]$

a) Construct Belief State & Apply Modus Ponens:

$KB = \{V_{21}, S_{11}, B_{22}\}$

rule 1 : Vibration $[2, 1] \rightarrow$ Mine adj $[2, 1]$

we have : Vibration $[2, 1]$

By modus Ponens: Mine Adj $[2, 1]$

rule 2 : smoke $[1, 1] \rightarrow$ Fire adj $[1, 1]$

we have : smoke $[1, 1]$, By modus Ponens : Fire adj $[1, 1]$

rule 3: Beacon $[x, y] \rightarrow$ Safe Zone Marker $[x, y]$

Sub, $x = 2, y = 2 \rightarrow$ Beacon $[2, 2] \rightarrow$ Safe Zone Marker $[2, 2]$

By modulus Powers: Safe Zone Marker $[2, 2]$

b) Forward chaining:

Step 1: Vibration $[2, 1] \rightarrow$ Mine adj $[2, 1]$

Mine adj $[2, 1]$

Step 2: Smoke $[1, 1] \rightarrow$ Fire adj $[1, 1]$

Fire adj $[1, 1]$

Step 3: Beacon $[2, 2] \rightarrow$ Safe Zone Marker $[2, 2]$

Safe Zone Marker $[2, 2]$

c) Is the Path Safe?

Path: $[1, 1] \rightarrow [2, 1] \rightarrow [2, 2]$

Cell $[1, 1]$, the robot perceives: smoke $[1, 1]$

$\therefore$ fire adj $[1, 1]$

Cell $[2, 1]$, the robot perceives: vibration $[2, 1]$

$\therefore$ mine adj $[2, 1]$

Cell $[2, 2]$, Beacon $[2, 2]$

$\therefore$ Safe Zone Marker $[2, 2]$

However, the rule for safety is: Mine $[x, y] \wedge$ Fire $[x, y] \rightarrow$ Safe $[x, y]$

To prove a cell is safe, the robot needs both:

Mine $[x, y]$ and Fire $[x, y]$

Conclusion: The path cannot be logically confirmed as safe.

The path is not proven safe: therefore, a logical agent should treat it as unsafe/uncertain and avoid moving until additional information is obtained.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks (100)
Understanding of Case	25	Thorough understanding, clearly identifies all key issues	Good understanding with most issues identified	Basic understanding, some issues missed	Poor understanding, problem not identified correctly	24
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	24
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis, lacks reasoning	19
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	19
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity, difficult to understand	9

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